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① No behavioural or causal claims, just accounting where $V \equiv \frac{PY}{M}$ so it's always true

↳ Also no directionality between relations

② Baumol-Tobin: $\frac{M}{P} = \left(\frac{YF}{2i} \right)^{1/2} = \frac{M^D}{P}$

$M.V = P.Y \rightarrow \frac{M}{P} = \frac{1}{V} \cdot Y \sim \frac{1}{V} = f(\bar{F}, \bar{Y}, \bar{i})$

connected

more specifically: $V = \left(\frac{2i \cdot Y}{F} \right)^{1/2}$

velocity negatively related to demand for liquidity

③ $M^S \uparrow + \bar{P} \rightarrow \underbrace{Y \uparrow \text{ or } i \downarrow}_{\text{combination}}$

otherwise P can change

→ absorb ΔM^S instead of m^d

→ No guarantee of any effect/change

(d) $M^s \rightarrow$ will allow i to vary such that

$$M^s = \frac{M^D}{P}$$

If they fix $i \rightarrow$ will have to allow M^s to vary to keep equilibrium

(2) $M^s = P \cdot m^D(Y, i)$

$$\frac{dM^s}{dt} = \frac{dP}{dt} \cdot m^D(Y, i) + P \cdot \frac{dm^D(Y, i)}{dt}$$

$$* \frac{dM^D}{dt} = \frac{\partial m^D}{\partial Y} \cdot \frac{dY}{dt} + \frac{\partial m^D}{\partial i} \cdot \frac{di}{dt}$$

* $\rightarrow 0$

$$\therefore \frac{\frac{dM^s}{dt}}{M^s} = \frac{\frac{\partial m^D}{\partial Y} \cdot \frac{dY}{dt} \cdot P}{P \cdot m^D(Y, i)} + \frac{\frac{dP}{dt} \cdot m^D(Y, i)}{P \cdot m^D(Y, i)}$$

$\downarrow$
 μ

π

$$\mu = \pi + \frac{\frac{\partial m^D}{\partial Y} \cdot \frac{dY}{dt} \cdot Y}{m^D(Y, i)} \rightarrow \mu = \pi + \frac{\frac{\partial m^D}{\partial Y} \cdot Y}{m^D} \cdot g$$

η

g

$$\therefore M = \pi + \eta \cdot g \rightarrow \pi = M - \eta \cdot g$$

• Inflation comes from difference money supply growth and real output/income growth

$$b) m^D = \left(\frac{YF}{2i} \right)^{1/2}$$

$$\frac{\partial m^D}{\partial Y} = \frac{1}{2} \cdot \left(\frac{YF}{2i} \right)^{-1/2} \cdot \frac{F}{2i}$$

$$\frac{\partial m^D}{\partial Y} \cdot \frac{Y}{m^D} = \frac{1}{2} \cdot \left(\frac{YF}{2i} \right)^{-1/2} \cdot \frac{F}{2i} \cdot \frac{Y}{\left(\frac{YF}{2i} \right)^{1/2}} \rightarrow \frac{1}{2} \cdot \frac{F}{2i} \cdot \frac{Y}{\frac{YF}{2i}} = \boxed{\frac{1}{2}} = \eta$$

$$1,02 = M - 0,5 \cdot 1,03 \rightarrow M = 1,035 \approx 3,5\% \text{ a.a}$$

$$\pi = 1,035 - 1,03 \rightarrow 0,5\%$$

© Key is in deriv. of $\frac{dm^D(p, y)}{dt}$:

$$M^S = P \cdot m^D(y, i)$$

$$\frac{dM^S}{dt} = \frac{dP}{dt} \cdot m^D(y, i) + P \cdot \frac{dm^D(y, i)}{dt}$$

*

was constant before

$$* \frac{dm^D}{dt} = \frac{\partial m}{\partial y} \cdot \frac{dy}{dt} + \frac{\partial m}{\partial i} \cdot \frac{di}{dt} + \frac{\partial m}{\partial F} \cdot \frac{dF}{dt}$$

$$\frac{\frac{dM^S}{dt}}{M^S} = \frac{\frac{dP}{dt} \cdot m^D(y, i)}{P \cdot m^D(y, i)} + \frac{P \left[\frac{\partial m}{\partial y} \cdot \frac{dy}{dt} + \frac{\partial m}{\partial F} \cdot \frac{dF}{dt} \right]}{P \cdot m^D(y, i)}$$

$$\mu = \pi + \eta(g+f) \rightarrow \pi = \mu - \frac{1}{2}(g+f)$$

$\hookrightarrow \eta = \frac{1}{2}$

Falling f is only compatible with constant growing money supply if cost of holding money increases, as there are no changes in $i \rightarrow \pi \uparrow$